Solution

Exercises: Discrete-Time Convolution

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1. Graphical evaluation of convolution is used to find the (output, input) for a system given the (output, input) and the impulse response.

2. We use graphical evaluation of convolution to (develop insight regarding system behavior, obtain the system output faster than a computer).

3. Which of the following steps are involved in graphical evaluation of discrete-time convolution $y[n] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$. Select all that apply.

a) Find $x[n-k]$ as a function of k for some fixed value n

b) Find $x[n-k]h[k]$ as a function of n for some fixed value k

c) Sum all the values in $x[n-k]h[k]$ over k for some fixed value n

d) Find $x[n-k]$ as a function of n for some fixed value k

e) Find $x[n-k]h[k]$ as a function of k for some fixed value n

f) Sum all the values in $x[n-k]h[k]$ over n for some fixed value k

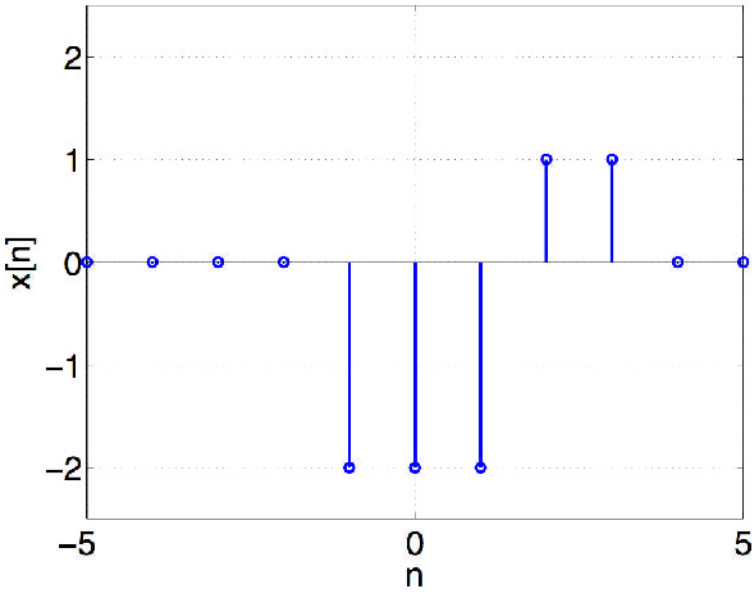
4. In order to find $x[n-k]$ for $n > 0$ you:

a) Reflect $x[k]$ about $k = 0$ to obtain $x[-k]$, then shift $x[-k]$ to the right for n steps

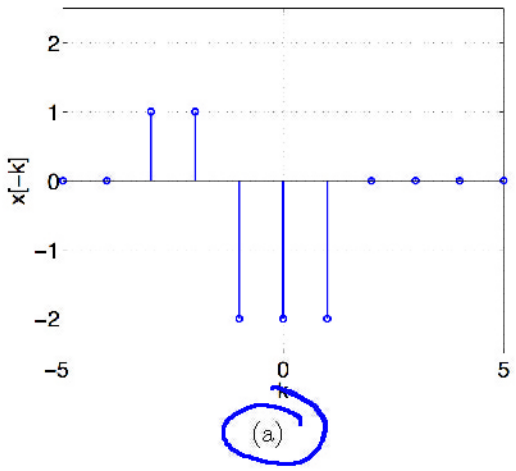
b) Reflect $x[k]$ about $k = 0$ to obtain $x[-k]$, then shift $x[-k]$ to the left for n steps

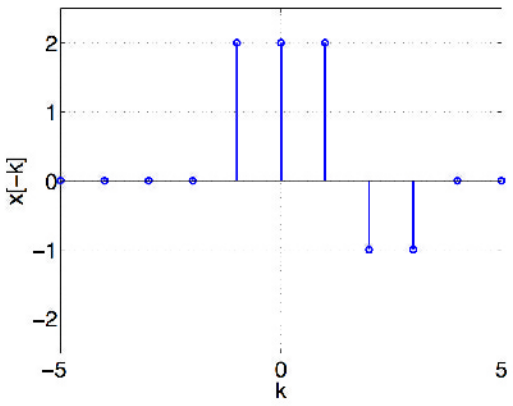
c) Shift $x[k]$ to the right for n steps, then reflect about $k = 0$

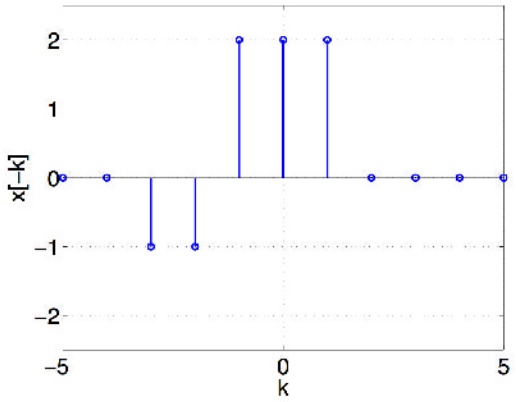
The discrete-time signal $x[n]$ is depicted in the figure below will be used in the next three problems.

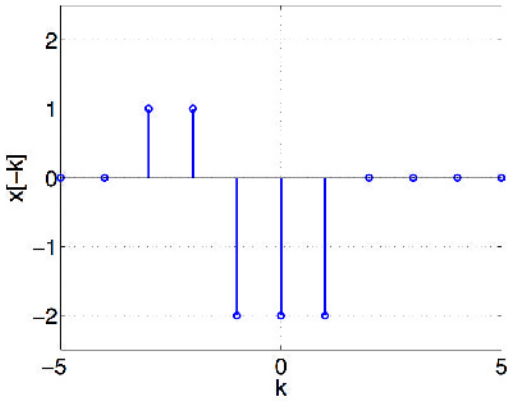


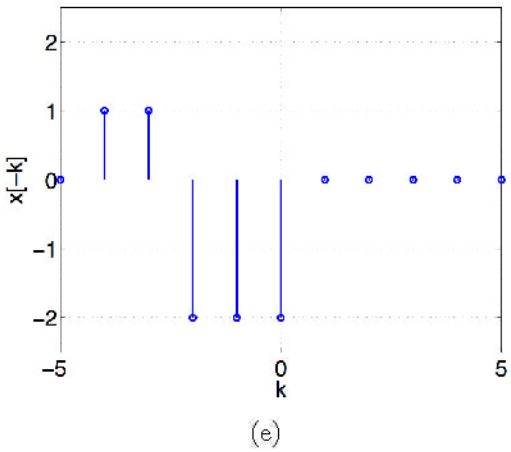
5. Select the answer corresponding to $x[-k]$.



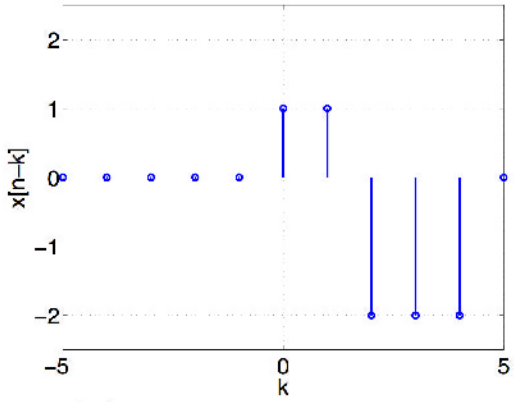






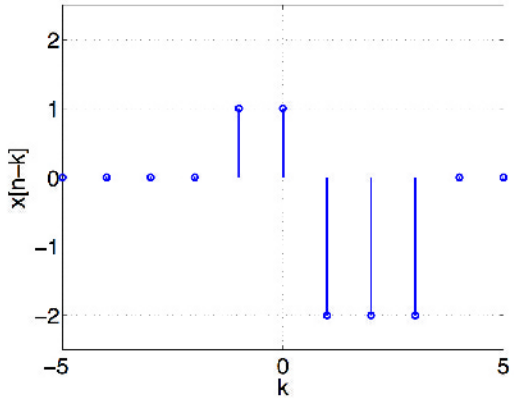


6. Each of the following graphs depict $x[n-k]$ for some value of n. Find the n corresponding to each graph.



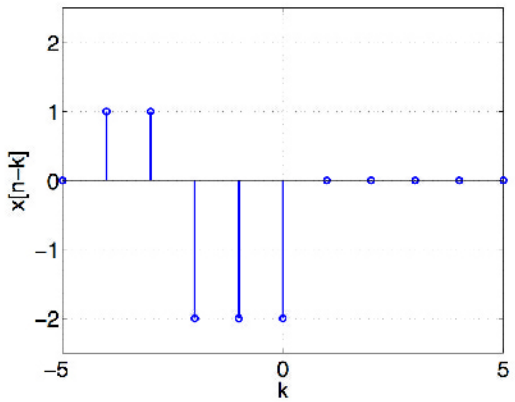
(f)

$n=3$



(g)

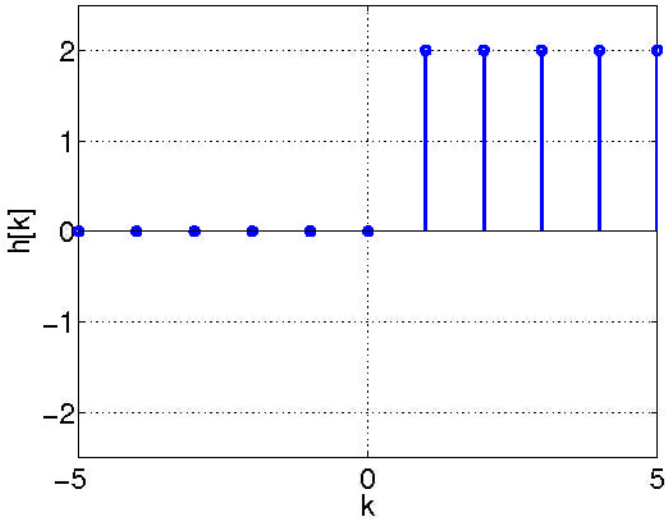
$n=2$



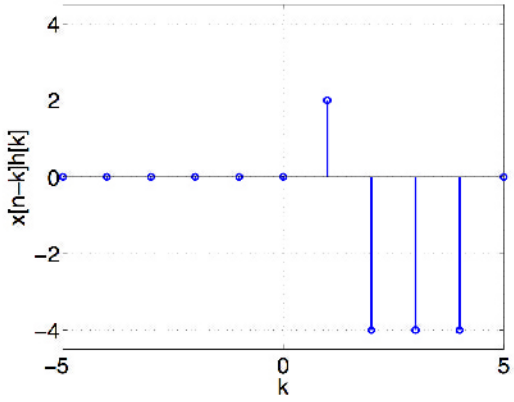
(h)

$n=-1$

7. Find the output of a system $y[n] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$ for $x[n]$ given previously. The impulse response of the system is $h[n] = 2u[n-1]$ as shown below.

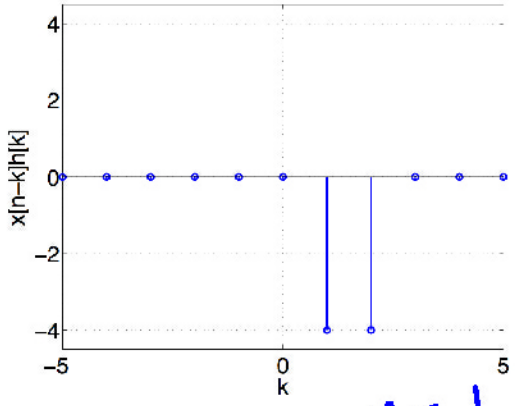


Each of the following graphs depict $x[n-k]h[k]$ for some value of n. Enter the n corresponding to each graph.



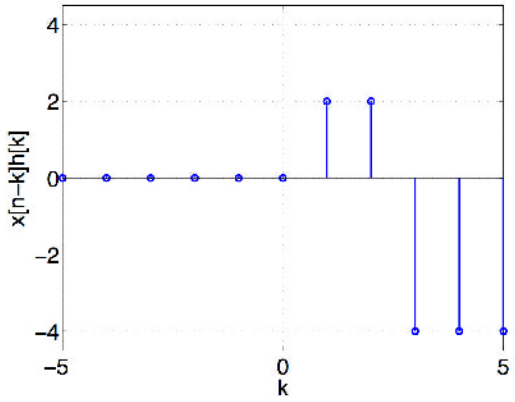
(i)

$n=3$



(j)

$n=1$

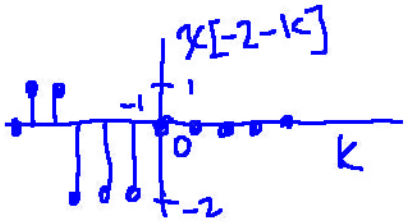


(k)

$n=4$

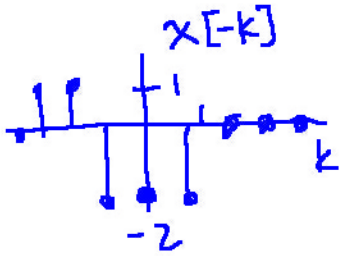
Now find the output $y[n]$ for the following values of n.

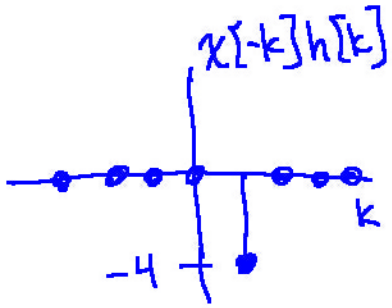
i) $y[-2]$



$x[-2-k]h[k] = 0 \Rightarrow y[-2] = 0$

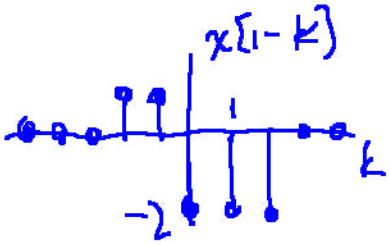
ii) $y[0]$

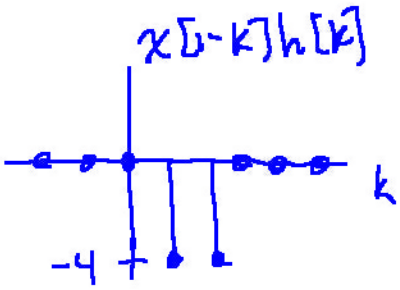




$y[0] = -4$

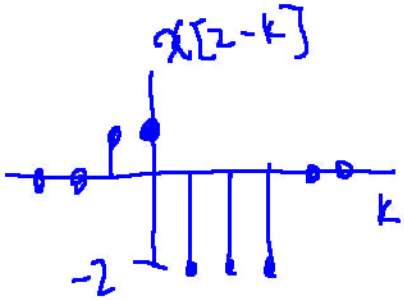
iii) $y[1]$

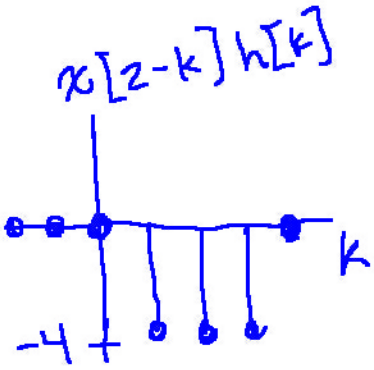




$y[1] = -4 + -4 = -8$

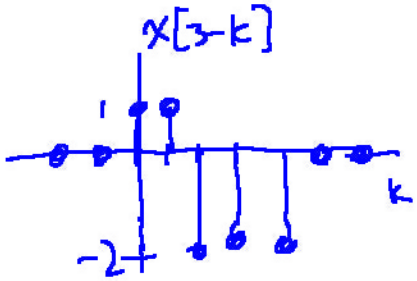
iv) $y[2]$

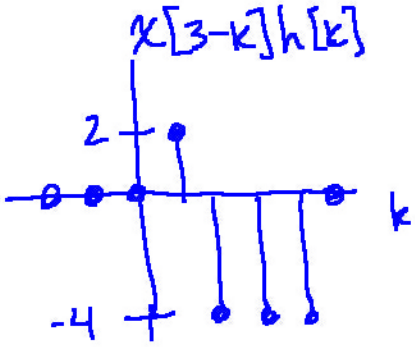




$y[2] = -4 + -4 + -4$
$= -12$

v) $y[3]$

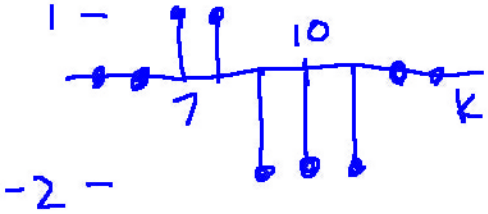


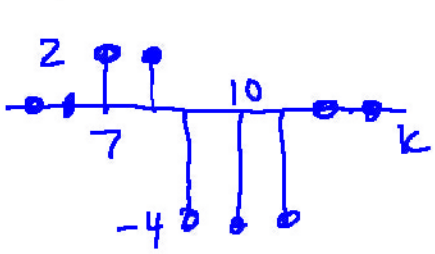


$y[3] = 2 - 4 - 4 - 4$

$= -10$

vi) $y[10]$





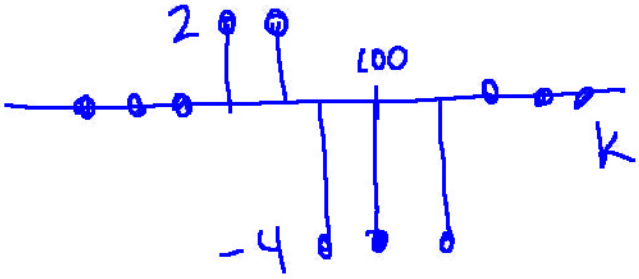
$y[10] = 2 + 2 - 4$

$-4 - 4$

$= -8$

vii) $y[100]$

$x[100-k]h[k]$

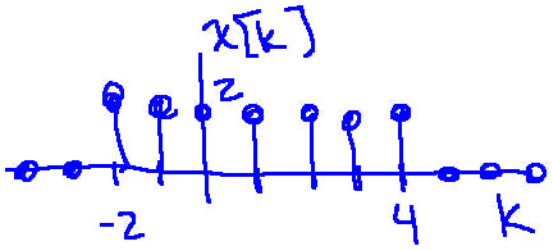


$y[100] = 2 + 2 - 4 - 4 - 4$

$= -8$

8. In this problem we compute $y[n] = x[n] * h[n]$ where $h[n] = (3/4)^n u[n]$ and $x[n] = 2(u[n+2] - u[n-5])$.

a) Graph $x[k]$. Identify the value of $x[k]$ on the following intervals.

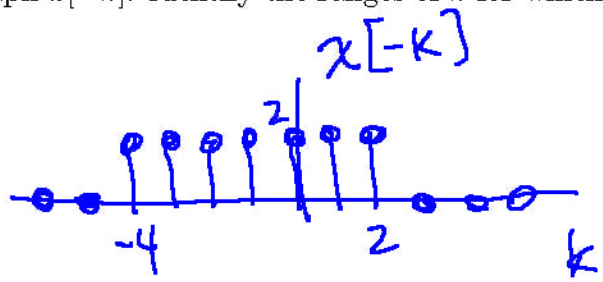


i) $x[k], k < -2$ → 0

ii) $x[k], -2 \leq k \leq 4$ → 2

iii) $x[k], k > 4$ → 0

b) Graph $x[-k]$. Identify the ranges of k for which $x[-k]$ takes the following values.

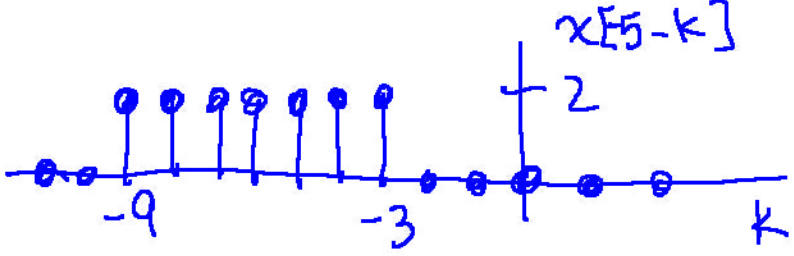


i) $x[-k] = 0$ for $k <$ -4

ii) $x[-k] = 2$ for -4 $\leq k \leq$ 2

iii) $x[-k] = 0$ for $k >$ 2

c) Graph $x[n-k]$ for $n = -5$. Enter the ranges of k for which $x[n-k]$ takes the following values:

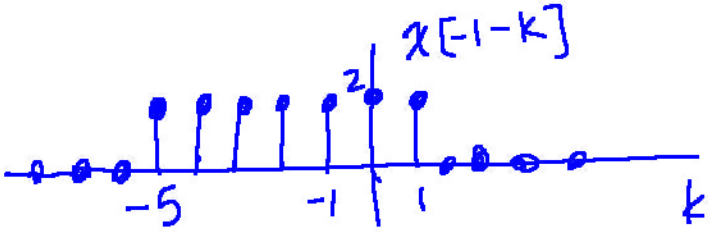


i) $x[n-k] = 0$ for $k <$ -9

ii) $x[n-k] = 2$ for -9 $\leq k \leq$ -3

iii) $x[n-k] = 0$ for $k >$ -3

d) Graph $x[n-k]$ for $n = -1$. Enter the ranges of k for which $x[n-k]$ takes the



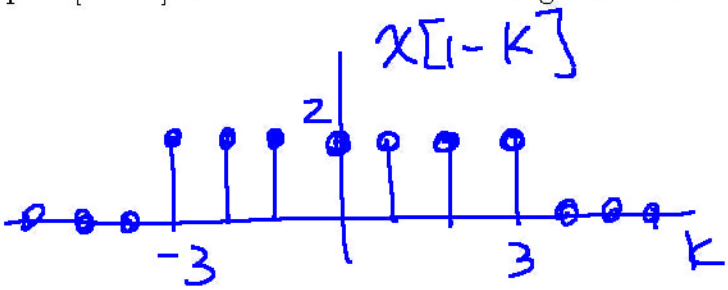
following values:

i) $x[n-k] = 0$ for $k <$ -5

ii) $x[n-k] = 2$ for -5 $\leq k \leq$ -1

iii) $x[n-k] = 0$ for $k >$ -1

e) Graph $x[n-k]$ for $n = 1$. Enter the ranges of k for which $x[n-k]$ takes the



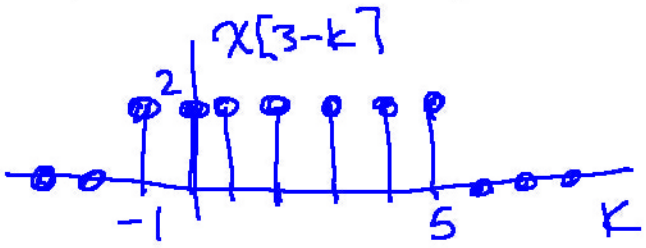
following values:

i) $x[n-k] = 0$ for $k <$ -3

ii) $x[n-k] = 2$ for -3 $\leq k \leq$ 3

iii) $x[n-k] = 0$ for $k >$ 3

f) Graph $x[n-k]$ for $n = 3$. Enter the ranges of k for which $x[n-k]$ takes the



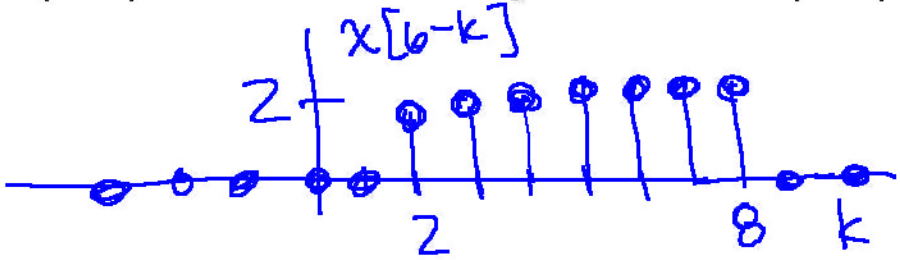
following values:

i) $x[n-k] = 0$ for $k <$ -1

ii) $x[n-k] = 2$ for -1 $\leq k \leq$ 5

iii) $x[n-k] = 0$ for $k >$ 5

g) Graph $x[n-k]$ for $n = 6$. Enter the ranges of k for which $x[n-k]$ takes the



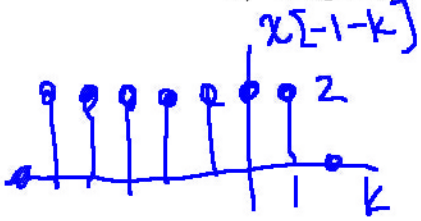
following values:

i) $x[n-k] = 0$ for $k <$ 2

ii) $x[n-k] = 2$ for 2 $\leq k \leq$ 8

iii) $x[n-k] = 0$ for $k >$ 8

h) Graph $x[n-k]h[k]$ for $n = -1$. Enter the ranges of k for which $x[n-k]h[k]$ takes





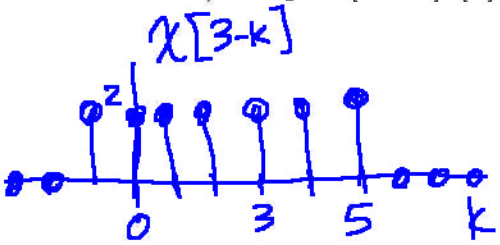
the following values:

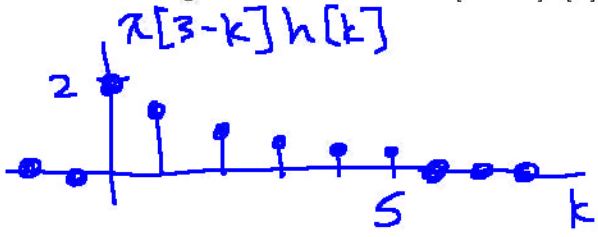
i) $x[n-k]h[k] = 0$ for $k <$ 0

ii) $x[n-k]h[k] = 2(3/4)^k$ for 0 $\leq k \leq$ 1

iii) $x[n-k]h[k] = 0$ for $k >$ 1

i) Graph $x[n-k]h[k]$ for $n = 3$. Enter the ranges of k for which $x[n-k]h[k]$ takes





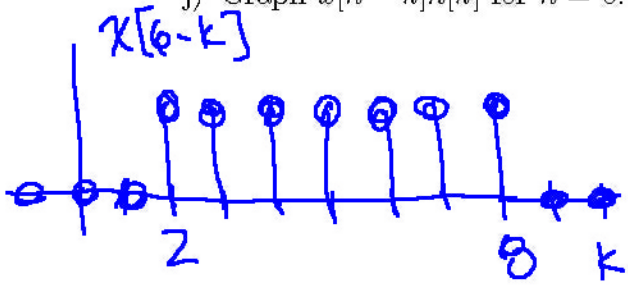
the following values:

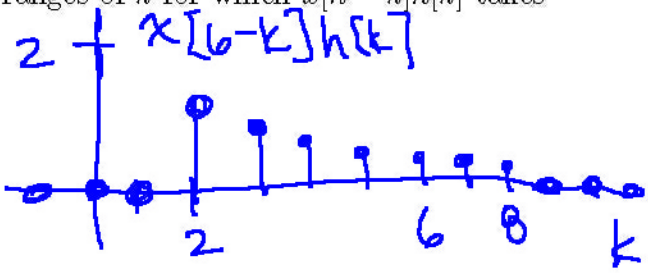
i) $x[n-k]h[k] = 0$ for $k <$ 0

ii) $x[n-k]h[k] = 2(3/4)^k$ for 0 $\leq k \leq$ 5

iii) $x[n-k]h[k] = 0$ for $k >$ 5

j) Graph $x[n-k]h[k]$ for $n = 6$. Enter the ranges of k for which $x[n-k]h[k]$ takes





the following values:

i) $x[n-k]h[k] = 0$ for $k <$ 2

ii) $x[n-k]h[k] = 2(3/4)^k$ for 2 $\leq k \leq$ 8

iii) $x[n-k]h[k] = 0$ for $k >$ 8

k) Now find $y[n]$ for the following values of n.

Hint: you may find it helpful to use the formula for a finite geometric series

$$\sum_{k=K_0}^{K_1} a^k = a^{K_0}\left(\frac{1-a^{K_1-K_0+1}}{1-a}\right)$$

for $a = 3/4$. This formula applies for all $a \neq 0$. Note that if K_0 is close to K_1 then it is probably simpler to add up the sum rather than use this closed form expression.

$y[-1] = 2 + 3/4 \cdot 2 = 3.5$

$y[3] = \sum_{k=0}^{5} 2(3/4)^k = 2\,\frac{1-(3/4)^6}{1-3/4} = 8\left(1-(3/4)^6\right) \approx 6.576$

$y[6] = \sum_{k=2}^{8} 2(3/4)^k = 2(3/4)^2\left(\frac{1-(3/4)^7}{1-3/4}\right)$

$= 8\left[(3/4)^2 - (3/4)^9\right]$

≈ 3.9

i) $y[-1] =$ 3.5

ii) $y[3] =$ 6.576

iii) $y[6] =$ 3.9

l) The output is zero for $n <$ -2

m) As $n \to \infty$ the output $y[n] \to$ 0

n) As n increases, the output initially (decreases, increases).

from $n = -3$